Quantitative Evaluation of Braitenberg-Based Pursuit Behaviour in Cluttered Surveillance Environments

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Abstract

Reactive control architectures inspired by Braitenberg vehicles offer compelling properties for surveillance robotics: low-latency response, minimal computational overhead, and no requirement for global maps or pre-built environment models. This paper presents a quantitative study of a Braitenberg-based pursuit controller deployed in simulated security environments with systematically varied obstacle densities. A differential-drive robot equipped with eight proximity sensors and a wide-angle target sensor pursues a mobile intruder using a superposition of an excitatory-crossed pursuit channel (Vehicle 2b) and an inhibitory-crossed avoidance channel (Vehicle 3a). We evaluate performance across eight obstacle density levels (0%–40% floor-area coverage) using four fixed layouts and five independent trials per layout, measuring capture rate, capture time, path efficiency, collision frequency, and heading change rate. We then introduce a four-stage pipeline: (i) the pure reactive baseline, (ii) a hybrid extension in which the pursuit and avoidance gains (α, β) are dynamically adapted each step by a lightweight reinforcement-learning policy trained with REINFORCE with baseline, (iii) an improved variant that adds cosine-weighted proximity blending, a forward bias, a mean-centred motor floor, and stuck-recovery to the base controller before applying the same learned weight adaptation, and (iv) an ablation of Stage 3 using the improved base kinematics with fixed weights and no RL, isolating the contribution of each component. All four architectures are compared across identical experimental conditions. Results show that the pure Braitenberg controller maintains 100% capture rate at low density but degrades sharply above 10%, reaching 0% at 25%. The Stage 2 hybrid RL controller performs comparably to or slightly below the baseline, indicating that weight adaptation alone on the original controller provides limited benefit. Results show that the pure Braitenberg controller maintains 100% capture rate at low density (mean 955 steps at 0%) but degrades sharply above 10%, reaching 0% by 25%. The Stage 2 hybrid RL controller performs comparably to or below the baseline at every density, confirming that weight adaptation alone provides no benefit on the original base. Stage 4 (improved base, fixed weights) matched or exceeded Stage 3 (improved base with RL) on capture rate at every density — 90% vs. 80% at 15%, 55% at 20%, and 40% vs. 20% at 25% — while the RL policy's sole measurable contribution was a reduction in collision steps at high density (199 vs. 615 steps at 40%). These results demonstrate that the kinematic improvements are both necessary and sufficient for the performance gain, and that RL weight adaptation, as implemented here, provides no benefit to capture rate on the improved base.

1. Introduction

1.1 Motivation and Problem Statement

Modern security and surveillance applications increasingly demand autonomous robotic sys-

tems that can detect and pursue intruders in real-world environments without the infrastructure overhead of centralised monitoring or pre-mapped spaces. Cluttered environments — offices, warehouses, and public spaces with furniture, partitions, and irregular layouts — present a particular challenge: the robot must balance aggressive pursuit of a moving target against the need to navigate around obstacles reactively, without the latency that deliberative planning would introduce.

Braitenberg vehicles [1] offer a principled framework for this class of problems. By coupling sensors directly to actuators through fixed or learned weights, they produce behaviour that is fast, interpretable, and robust to sensor noise. However, fixed-weight reactive controllers are, by definition, static: the same gain values that produce efficient pursuit in an open arena may cause stalling, oscillation, or repeated collisions in a densely cluttered one.

This motivates our core research question:

> *How does obstacle density affect the performance of a reactive Braitenberg pursuit controller in cluttered security environments, and to what extent can dynamically adapted channel weights recover lost performance?*

1.2 Research Objectives

This work pursues three specific objectives:

i. Characterise the quantitative relationship between obstacle density and pursuit performance for a fixed-weight Braitenberg controller.

ii. Identify the failure modes — stalling, thrashing, and detection loss — that cause performance degradation at high density.

iii. Evaluate whether a three-stage hybrid architecture — in which improved control heuristics and a reinforcement-learning policy that adaptively tunes the pursuit gain α and avoidance gain β in real time — can recover performance without sacrificing the low-latency, reactive character of the base controller.

iv. Disentangle the respective contributions of the improved kinematic heuristics and the RL weight adaptation via a fourth ablation condition that applies the improved base with fixed weights only.

1.3 Scope and Limitations

All experiments are conducted in a 2D top-down simulation using a custom Python/Pygame environment. The 2D top-down abstraction is standard practice for differential-drive pursuit studies [10] and is justified here on three grounds: the robot operates on a flat floor with no terrain variation; all relevant sensing (proximity and target detection) occurs in the horizontal plane; and vertical dynamics have no bearing on the pursuit-avoidance behaviour under study. The robot and intruder are modelled as circular agents with differential-drive and constant-velocity (bouncing) kinematics respectively. Physical effects including sensor noise, actuator latency, and dynamic obstacles beyond the intruder are outside the scope of this work. The study uses only circular static obstacles because they keep the ray-casting geometry analytically tractable (ray-circle intersection has a closed-form solution) and represent a physically meaningful class of real-world features: building columns, support pillars, waste bins, and standalone furniture items all present approximately circular cross-sections in a top-down view. Non-circular obstacles such as walls, shelving units, and desks are a natural future extension.

2. Background and Related Work

2.1 Braitenberg Vehicles

Valentino Braitenberg's 1986 thought-experiment catalogue [1] described a family of simple vehicles whose behaviour — fear, aggression, love, curiosity — emerges entirely from the topology and sign of sensor-to-motor connections, without any explicit representation of the world. Vehicle 2 (excitatory connections, direct or crossed) produces approach or avoidance behaviour toward a stimulus source. Vehicle 3 (inhibitory connections) generates

speed modulation that can approximate more complex response patterns. The key insight is that apparent purposiveness arises from coupling, not from planning.

In robotics, Braitenberg-inspired architectures have been applied to obstacle avoidance [5, 6], light-tracking, and swarm coordination [7]. Their practical advantages — $O(n)$ computation in the number of sensors, no map requirement, and sub-millisecond decision cycles — make them attractive for platforms with constrained compute budgets.

2.2 Reactive Navigation in Cluttered Spaces

Pure reactive controllers face a well-documented challenge in cluttered environments: the sensor-to-motor coupling that produces smooth pursuit in free space can generate limit cycles or deadlock configurations when obstacles block the direct path to the target [8]. Several strategies have been proposed to address this, including potential-field blending [2], vector field histograms [3], and hybrid deliberative-reactive architectures. The present work takes a different approach: rather than adding a deliberative layer, we augment the weight parameters of the existing reactive controller using a learned policy, preserving the low-latency character while adapting to environmental conditions.

2.3 Learning-Augmented Reactive Controllers

The combination of evolutionary or learning methods with Braitenberg-style controllers has a modest but relevant literature. Floreano and Mondada [4] demonstrated that evolutionary algorithms could discover effective sensor-to-motor weight matrices for navigation tasks. More recently, reinforcement learning has been used to adapt the parameters of hand-designed controllers in real time [9], an approach sometimes termed *meta-control*. Our hybrid architecture is most closely related to this second category: the Braitenberg topology is fixed and interpretable, while a small RL policy outputs gain adjustments $\Delta\alpha$ and $\Delta\beta$ conditioned on the current sensor state.

3. System Architecture

3.1 Simulation Environment

The simulation arena is an 800×800 pixel bounded square with fully reflective walls, representing an approximately 8×8 m surveillance space such as a building lobby or open-plan office floor (scale: 1 px $\approx$ 1 cm).

Platform reference. Agent dimensions and sensor parameters are grounded in the TurtleBot3 Burger [10], a widely used differential-drive research platform with a roughly circular footprint ($\sim$138 mm diameter), maximum linear speed of 0.22 m/s, maximum angular speed of 2.84 rad/s, and a 160 mm track width. The e-puck robot [11] is additionally used as a reference for the proximity sensor ring, as it was designed specifically for Braitenberg vehicle experiments and carries eight IR sensors in the same evenly spaced configuration modelled here. The simulation is deliberately schematic rather than to-scale with any single platform: exact metre-per-second correspondence is not claimed, and the study focuses on the relative performance of control architectures across density conditions rather than absolute speed fidelity.

Agent geometry. All agents are modelled as circles. The robot has radius $r_{\text{rob}} = 10$ px ($\approx$20 cm diameter), representing a compact mobile robot footprint consistent with the TurtleBot3 class. The intruder has radius $r_{\text{int}} = 15$ px ($\approx$30 cm), representing the approximate horizontal cross-section of a standing adult as viewed from above in a top-down schematic. Obstacle radii are sampled uniformly from $[15, 30]$ px ($\approx$30–60 cm), representing columns, support pillars, and standalone furniture.

Kinematics and capture. The robot's maximum speed is $v_{\max} = 2.0$ px/step. The intruder moves at constant speed $v_{\text{int}} = 1.5$ px/step and reflects off walls and obstacles using angle-of-incidence reflection; obstacle contacts are resolved via the collision normal. The resulting speed ratio of 1.33$\times$ represents a scenario in

which the robot is faster than a cautiously moving intruder (a person moving slowly so as not to attract attention, at approximately 0.3–0.5 m/s), and is consistent with faster wheeled platforms such as the Pioneer 3-AT or Robotino operating in the same space. Capture is declared when the Euclidean distance between robot and intruder centres falls below $r_{\text{capture}} + r_{\text{int}} = 40\,\text{px}$ ($r_{\text{capture}} = 25\,\text{px}$), representing the robot coming within approximately 50 cm of the intruder.

Obstacle density levels. Obstacle density ρ is defined as the ratio of total obstacle area to arena area (Equation 1). The eight levels evaluated span qualitatively distinct real-world scenarios:

Density	Real-world analogue
0%	Empty corridor or open lobby
5%	Sparsely furnished office
10%	Office with columns and furniture
15%	Mixed open/cluttered floor plan
20%	Cluttered warehouse floor
25%	Dense warehouse or factory floor
30%	Heavily obstructed industrial space
40%	Near-impassable (stress test)

$$\rho = \frac{\sum_i \pi r_i^2}{W \cdot H} \tag{1}$$

where $W = H = 800\,\text{px}$.

3.2 Stage 1: Pure Braitenberg Controller

The robot carries two sensor modalities:

Proximity sensors. Eight sensors are distributed evenly around the robot body at angular intervals of $2\pi/8 = 45$ř in the robot frame, matching the IR sensor ring of the e-puck platform [11]. Each casts a ray of maximum range $d_{\text{prox}} = 60\,\text{px}$ (≈60 cm), consistent with the detection range of short-range infrared proximity sensors, and returns a normalised reading $p_i \in [0, 1]$:

$$p_i = 1 - \frac{d_i}{d_{\text{prox}}} \tag{2}$$

where d_i is the distance to the nearest obstacle or wall along that ray. A reading of $p_i = 0$ indicates a clear path to the sensor's maximum range; $p_i = 1$ indicates contact.

Target sensor. A pair of wide-angle sensors (field-of-view half-angle $\phi = 1.5\,\text{rad}$, approximately 86° per side, 172° total) detect the intruder within range $d_{\text{tgt}} = 600\,\text{px}$ (≈6 m), representative of a passive infrared (PIR) motion sensor or wide-angle camera mounted on the robot. The 6 m detection range is intentionally less than the arena diagonal (≈11.3 m), so the intruder can genuinely exit the robot's sensing envelope at high obstacle densities, making initial detection a meaningful challenge. For a target at distance d_t and bearing θ relative to the robot heading, the active sensor intensity is:

$$s_L = \left(1 - \frac{d_t}{d_{\text{tgt}}}\right) \cos\left(\frac{|\theta|}{\phi} \cdot \frac{\pi}{2}\right), \quad \theta \geq 0 \tag{3}$$

and s_R is defined symmetrically for $\theta < 0$. Both readings are zero when $|\theta| > \phi$ or $d_t > d_{\text{tgt}}$, where $\phi = 1.5\,\text{rad}$.

Motor command computation. The two Braitenberg channels are superimposed additively to produce the final left and right motor commands.

The *pursuit channel* implements Vehicle 2b (excitatory, crossed): each motor speed is driven in proportion to the *opposite* target sensor, causing the robot to steer toward the stimulus source:

$$v_L^{\text{pursuit}} = \alpha \cdot g_t \cdot s_R \tag{4}$$

$$v_R^{\text{pursuit}} = \alpha \cdot g_t \cdot s_L \tag{5}$$

where $\alpha = 1.0$ is the pursuit weight and $g_t = 2.5$ is a fixed target gain.

The *avoidance channel* implements Vehicle 3a (inhibitory, crossed): proximity readings are aggregated separately for the left and right body halves by taking the *maximum* reading within each hemisphere:

$$P_L = \max_{i \in \mathcal{L}} p_i \tag{6}$$

$$P_R = \max_{i \in \mathcal{R}} p_i \tag{7}$$

where $\mathcal{L}$ and $\mathcal{R}$ are the index sets of sensors in the left and right body hemispheres (first and second halves of the evenly spaced array, respectively). A strong reading on the left side

inhibits the *right* motor, turning the robot away from the obstacle:

$$v_L = v_L^{\text{pursuit}} - \beta \cdot g_p \cdot P_R \tag{8}$$
$$v_R = v_R^{\text{pursuit}} - \beta \cdot g_p \cdot P_L \tag{9}$$

where $\beta = 1.2$ is the avoidance weight and $g_p = 1.5$ is a fixed proximity gain. Final motor speeds are clamped to $[0, v_{\max}]$ where $v_{\max} = 2.0\,\text{px/step}$; negative values are clipped to zero. No forward bias, motor floor, or stuck-recovery heuristics are applied: the controller is a pure superposition of the two Braitenberg channels.

3.3 Stage 3: Improved Braitenberg Controller

To address known failure modes of the pure reactive controller at high obstacle densities, an improved base controller (`BraitenbergRobotImproved`) introduces four modifications:

Cosine-weighted proximity aggregation. Rather than taking the hemisphere maximum, all eight sensors contribute to the avoidance load with a weight proportional to their forward-facing component:

$$w_i = \max(0,\, \cos\phi_i) + 0.3 \tag{10}$$

where ϕ_i is the sensor's angle in the robot frame. The 0.3 offset preserves a modest rearward contribution. Left-side and right-side loads are accumulated as weighted sums $\sum_{\mathcal{L}} p_i w_i$ and $\sum_{\mathcal{R}} p_i w_i$, then normalised by the number of sensors. This produces smoother avoidance responses compared to the discontinuous max-aggregation of Stage 1.

Forward bias. A constant bias of $b_f = 0.15\,\text{px/step}$ is added to both motor commands after the avoidance channel, ensuring the robot continues searching even when the intruder lies outside sensor range.

Motor floor. Rather than clamping each motor independently, the floor is applied to the mean motor speed: if $(v_L + v_R)/2 < v_{\text{floor}}$, both motors are raised by the deficit, preserving the net forward momentum at $v_{\text{floor}} = 0.2\,\text{px/step}$ while still allowing asymmetric motor commands (and therefore tight turns) when avoidance demands them. This prevents the avoidance channel from stalling the robot while retaining the ability to slow one side for obstacle avoidance.

Stuck detection and recovery. Every $P_{\text{stuck}} = 50$ steps, the robot checks whether it has displaced less than $\delta_{\text{stuck}} = 8\,\text{px}$ since the last checkpoint. If so, it enters a reverse-turn recovery manoeuvre lasting $D_{\text{rec}} = 15$ steps at reduced speed ($0.3 \times v_{\max}$, turn asymmetry factor 0.4), with the turn direction alternating between recoveries to avoid repeated deadlocks. The shorter, gentler recovery compared to earlier iterations reduces the number of collision steps incurred during escape.

3.4 Hybrid Learned-Weight Extension (Stages 2 and 3)

The hybrid controller retains the full Braitenberg architecture of the respective base (Stage 1 or Stage 3) but replaces the fixed scalar weights (α, β) with a policy π_θ that outputs weight adjustments $(\Delta\alpha, \Delta\beta)$ at each time step:

$$\alpha_t = \bar{\alpha} + \Delta\alpha_t, \qquad \Delta\alpha_t = \pi_\theta^{\alpha}(\mathbf{o}_t) \tag{11}$$
$$\beta_t = \bar{\beta} + \Delta\beta_t, \qquad \Delta\beta_t = \pi_\theta^{\beta}(\mathbf{o}_t) \tag{12}$$

where $\bar{\alpha} = 1.0$ and $\bar{\beta} = 1.2$ are the nominal (pure Braitenberg) values and $\mathbf{o}_t$ is the observation vector. Effective weights are bounded to $[\alpha_{\min}, \alpha_{\max}] = [\beta_{\min}, \beta_{\max}] = [0.1, 2.5]$.

Observation space. The policy observes:

$$\mathbf{o}_t = \left[p_1, \ldots, p_8,\ s_L,\ s_R,\ \hat{d}_t,\ \hat{\theta}_t\right] \tag{13}$$

where $\hat{d}_t = \min(d_t/d_{\text{tgt}},\, 1)$ and $\hat{\theta}_t = \theta_t/\pi$ are normalised distance and bearing to the intruder. The observation vector has dimension $|\mathbf{o}_t| = 12$.

Policy architecture. The policy is a shared-trunk actor-critic network:

$$\text{trunk} : \mathbb{R}^{12} \xrightarrow{W_1, b_1} \mathbb{R}^{32} \xrightarrow{\text{ReLU}} \mathbb{R}^{32} \tag{14}$$

$$\text{actor}: \quad \mu = \tanh(W_a \cdot h + b_a) \cdot \delta_{\max} \tag{15}$$

$$\text{critic}: \quad V = W_v \cdot h + b_v \tag{16}$$

with $W_1 \in \mathbb{R}^{32\times 12}$, $W_a \in \mathbb{R}^{2\times 32}$, $W_v \in \mathbb{R}^{1\times 32}$, and $\delta_{\max} = 0.5$. The actor outputs a Gaussian mean $\mu \in [-0.5, 0.5]^2$ with a learned log-standard deviation initialised to -0.5; actions are sampled stochastically during training and set to μ (greedy) during evaluation. All weights are initialised with Xavier (Glorot) initialisation.

Reward signal. The per-step reward is:

$$r_t = -\lambda_d \hat{d}_t - \lambda_c \mathbb{1}[\text{collided}] - \lambda_\tau + \lambda_{\text{cap}} \mathbb{1}[\text{captured}] \tag{17}$$

with $\lambda_d = 0.03$, $\lambda_c = 0.15$, $\lambda_\tau = 0.002$ (per-step time penalty), and $\lambda_{\text{cap}} = 15.0$. The time penalty encourages speed; the collision penalty discourages repeated obstacle contact; the large capture bonus provides a strong terminal signal. Episode length is capped at 4,000 steps during evaluation and 2,000 steps during training.

Training algorithm. The policy is trained using REINFORCE with baseline [**?**]. Each training episode collects a full trajectory under the stochastic policy. Discounted returns $G_t = \sum_{k\geq t} \gamma^{k-t} r_k$ are computed with discount factor $\gamma = 0.99$, and advantages are formed as $A_t = G_t - V(s_t)$ and normalised across the episode. Actor parameters are updated by gradient ascent on the expected return:

$$\nabla_\theta J \approx \frac{1}{T} \sum_{t=0}^{T-1} A_t \, \nabla_\theta \log \pi_\theta(a_t \mid s_t) \tag{18}$$

The critic is updated by gradient descent on the mean-squared Bellman error. All gradients are computed analytically in pure NumPy (no external ML framework) and clipped to a global norm of 1.0. Learning rates are $\eta_{\text{actor}} = 5 \times 10^{-4}$, $\eta_{\text{critic}} = 1 \times 10^{-3}$, and $\eta_{\log\sigma} = 1 \times 10^{-4}$. An entropy bonus with coefficient $\lambda_H = 0.01$ is added to the actor update to maintain exploration.

Curriculum. Training proceeds in two phases: for the first 100 episodes, only easy obstacle densities ($\rho \in \{0\%, 5\%, 10\%\}$) are sampled, allowing the policy to learn basic pursuit before encountering cluttered layouts. From episode 101 onward, the full density range (0%–40%) is used. The policy is trained for 2,000 episodes in total with checkpoints saved every 250 episodes. The same training procedure is applied independently to produce a Stage 2 policy (on the original Braitenberg base) and a Stage 3 policy (on the improved base).

4. Experimental Design

4.1 Pipeline Stages and Controllers

Four controllers are evaluated:

i. **Stage 1 (Baseline):** Pure Braitenberg controller with fixed gains $\alpha = 1.0$, $\beta = 1.2$ (Section 3.2).

ii. **Stage 2 (Hybrid):** Original Braitenberg base with RL-adapted gains (α_t, β_t) from a trained policy.

iii. **Stage 3 (Improved Hybrid):** Improved Braitenberg base (cosine blending, forward bias, motor floor, stuck recovery) with RL-adapted gains from a separately trained policy.

iv. **Stage 4 (Improved Baseline):** Improved Braitenberg base with fixed gains $\alpha = 1.0$, $\beta = 1.2$ and *no* RL policy. This ablation isolates the contribution of the kinematic improvements from the RL weight adaptation by holding the weights constant at their nominal values while using the improved base.

4.2 Density Levels and Layout Protocol

Eight obstacle density levels were evaluated: $\rho \in \{0\%, 5\%, 10\%, 15\%, 20\%, 25\%, 30\%, 40\%\}$. At each density level, four distinct obstacle layouts were generated deterministically from fixed random seeds derived from a base seed of 42. Five independent trials were conducted per layout (varying the intruder's initial heading via a per-trial seed), yielding $N = 20$ trials per density level and $N_{\text{total}} = 160$ trials per controller. All four controllers were evaluated on *identical* layouts and trial seeds to ensure a fair comparison.

Obstacles were placed by rejection sampling: a candidate circle was rejected if it overlapped any existing obstacle or fell within a clearance radius of 60 px of the robot start position or 50 px of the intruder start position.

4.3 Performance Metrics

Five metrics were computed per trial:

i. **Capture rate**: fraction of trials in which the robot captured the intruder before the timeout ($T_{\max} = 4000$ steps).

ii. **Capture time** t_c: number of simulation steps from trial start to capture. Trials that time out are assigned $t_c = T_{\max}$.

iii. **Path ratio** ρ_p: ratio of the robot's total travel distance to the straight-line distance from the robot's starting position to the intruder's position at the end of the trial:

$$\rho_p = \frac{L_{\text{path}}}{d(x_0, y_0,\, x_t^{\text{int}}, y_t^{\text{int}})} \tag{19}$$

Values close to 1 indicate a near-direct route; larger values indicate winding or oscillatory paths.

iv. **Collision count**: number of simulation steps in which the robot body overlapped an obstacle.

v. **Pursuit efficiency index** η: a composite metric penalising both slow capture and high collision count:

$$\eta = \frac{1}{t_c \cdot (1 + N_{\text{coll}}) + \epsilon} \tag{20}$$

where $\epsilon = 10^{-6}$ prevents division by zero.

Additionally, two diagnostic metrics were logged per trial: **time to first detection** (the step at which either target sensor first registered a non-zero reading, or -1 if the intruder was never detected) and **mean heading change rate** (mean absolute angular displacement per step, in rad/step), which quantifies directional stability and indirectly measures avoidance-channel thrashing.

4.4 Statistical Analysis

All metrics are reported as mean ± standard deviation across the 20 trials per density level. Differences between controllers are assessed using a two-sample Welch's t-test (significance threshold $p < 0.05$) applied independently at each density level. Effect sizes are reported using Cohen's d.

5. Results

5.1 Stage 1: Pure Braitenberg Controller

Table 1 summarises capture rates and mean capture times for all four controllers. The pure Braitenberg controller achieved 100% capture at 0% density (mean 955 steps) and 90% at 5% (mean 1,454 steps), demonstrating robust pursuit in sparse environments (Figure 1). Performance declined sharply thereafter: 60% at 10% (mean 2,679 steps), 35% at 15% (mean 3,277 steps), and 10% at 20% (mean 3,902 steps), before collapsing to 0% at 25% and above, where all trials timed out at 4,000 steps (Figure 2).

Collision counts for Stage 1 were non-monotonic: rising from 210 steps at 5% to a peak of 429 steps at 15%, then falling to 146 steps at 20% and near zero at 25%–40% (Figure 3). This counter-intuitive decrease at high density reflects the onset of stalling: a robot that has ceased meaningful movement incurs no further collisions.

5.2 Diagnostic Metrics

The pursuit efficiency index η for Stage 1 peaks at 0% density ($\eta \approx 0.00175$) and declines steeply through 5% ($\eta \approx 0.00090$) and 10% ($\eta \approx 0.00030$), before plateauing at a low but non-zero value ($\eta \approx 0.00017$–0.00025) from 20% onward (Figure 4). The plateau arises because timed-out trials contribute the maximum $t_c = 4{,}000$ steps with near-zero collision counts, producing a small but constant η. The non-monotonic collision peak at 15% coincides with the density at which the robot still attempts pursuit — and therefore still contacts obstacles — but struggles to navigate around them, representing the transition between the thrashing and stalling failure regimes.

5.3 Stage 2: Hybrid RL-Adapted Weights

The Stage 2 controller matched Stage 1 at 0% density (100%, mean 1,072 steps) but underperformed at 5% (80% vs. 90%, mean 1,892 vs. 1,454 steps) and declined more steeply at intermediate densities: 65% at 10% (mean 2,690 steps), 20% at 15% (mean 3,638 steps), and 5% at 20% (mean 3,952 steps) (Figure 1). Both Stage 1 and Stage 2 reached 0% capture from 25% onward.

Collision counts for Stage 2 were broadly similar to Stage 1 at low density (279 steps at 5%, 79 steps at 10%) but diverged at 15% (125 vs. 429 for Stage 1), then fell to near zero from 20% onward (5 steps at 20%, 24 at 25%, and 0 at 30%–40%) (Figure 3). This collapse in collision count at 20%+ is not evidence of improved avoidance: it reflects earlier onset of stalling relative to Stage 1, consistent with the lower success rate. The pursuit efficiency of Stage 2 tracked Stage 1 closely at all densities (Figure 4), confirming that RL weight adaptation provided no meaningful performance improvement on the original base.

5.4 Stage 3: Improved Hybrid Controller

Stage 3 substantially outperformed Stage 1 and Stage 2 from 10% density onward. At 0% and 5% density, Stage 3 matched Stage 1 at 100% capture with faster mean capture times (869 and 893 steps vs. 955 and 1,454 for Stage 1). At 10%, Stage 3 achieved 80% (mean 1,853 steps) compared to Stage 1's 60% (2,679 steps) and Stage 2's 65% (2,690 steps). At 15%, Stage 3 maintained 80% capture (mean 2,089 steps) while Stage 1 and Stage 2 had fallen to 35% and 20% respectively. This advantage continued at 20% (55%, mean 3,314 steps vs. 10%/5% for Stage 1/2) and 25% (20%, mean 3,509 steps vs. 0% for both). At 30% and 40%, Stage 3 also failed with all trials timing out (Figure 1).

Collision counts for Stage 3 increased monotonically with density: 51 steps at 5%, 337 at 10%, 303 at 15%, 508 at 20%, 496 at 25%, 415 at 30%, and 199 at 40% (Figure 3). The reduction at 40% reflects the onset of timeout-dominated trials where the robot spends most of its budget navigating unsuccessfully rather than contacting obstacles. These elevated collision counts reflect the sustained forward momentum from the motor floor and forward bias: the robot pushes through tight gaps rather than stalling, enabling more captures but at the cost of more obstacle contact.

Pursuit efficiency for Stage 3 was highest at 0% and 5% ($\eta \approx 0.00192$ and 0.00097), exceeding Stage 1 and Stage 2 at those levels due to faster capture times and low collision counts. From 10% onward, efficiency collapsed to near zero as the high collision counts in the denominator of η overwhelmed the improvement in capture time (Figure 4).

5.5 Stage 4: Improved Base with Fixed Weights (Ablation)

Stage 4 uses `BraitenbergRobotImproved` with fixed nominal gains ($\alpha = 1.0$, $\beta = 1.2$) and no RL policy, isolating the kinematic improvements from the weight adaptation.

Stage 4 captured at 100% for 0% and 5% density (mean 862 and 867 steps), 85% at 10% (mean 1,733 steps), 90% at 15% (mean 1,816 steps), 55% at 20% (mean 3,044 steps), and 40% at 25% (mean 3,078 steps), before collapsing to 0% at 30% and 40% (Figure 1). Notably, Stage 4 equalled or exceeded Stage 3 on capture rate at every density level: matching at 0%, 5%, 20%, and 25%; and outperforming by 5 percentage points at 10% (85% vs. 80%) and 10 percentage points at 15% (90% vs. 80%). Mean capture times were also equal to or faster than Stage 3 at all densities (Figure 2).

The one metric where Stage 3 outperforms Stage 4 is collision count. Stage 4 collisions were higher than Stage 3 at every density above 5%: 346 vs. 337 steps at 10%, 286 vs. 303 steps at 15%, 599 vs. 508 steps at 20%, 583 vs. 496 steps at 25%, 641 vs. 415 steps at 30%, and 615 vs. 199 steps at 40% (Figure 3). This divergence is most pronounced at 30%–40%, where Stage 3's RL collision penalty has the greatest effect relative to the fixed-weight Stage 4. Pursuit efficiency for Stage 4 closely matched Stage 3, with Stage 4 marginally higher at 0% and 5% due to its slightly faster capture times and lower collision counts in open environments (Figure 4).

Table 1: Capture rate (%) for all four pipeline stages across obstacle density levels ($N = 20$ trials each). Mean capture time (steps) shown for Stage 1 and Stage 4 to illustrate the effect of improved kinematics. Timeout = 4000 steps.

Density	Capture rate (%)				Mean step	
	S1	S2	S3	S4	S1	S4
0%	100	100	100	100	955	862
5%	90	80	100	100	1454	867
10%	60	65	80	85	2679	1733
15%	35	20	80	90	3277	1816
20%	10	5	55	55	3902	3044
25%	0	0	20	40	4000	3078
30%	0	0	0	0	4000	4000
40%	0	0	0	0	4000	4000

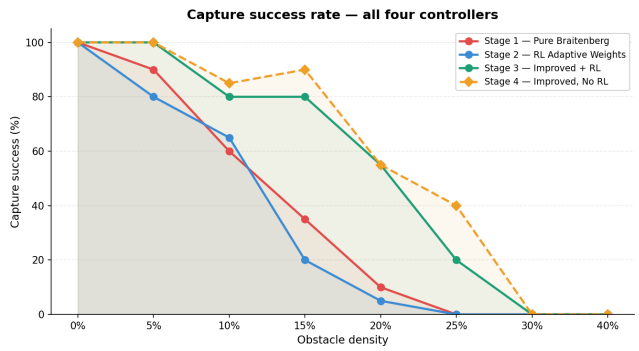

Figure 1: Capture success rate (%) vs. obstacle density for all four pipeline stages. Stage 3 (improved + RL) and Stage 4 (improved, no RL) curves are nearly indistinguishable, confirming that the kinematic improvements rather than the RL weight adaptation drive the performance gain over Stage 1 and Stage 2.

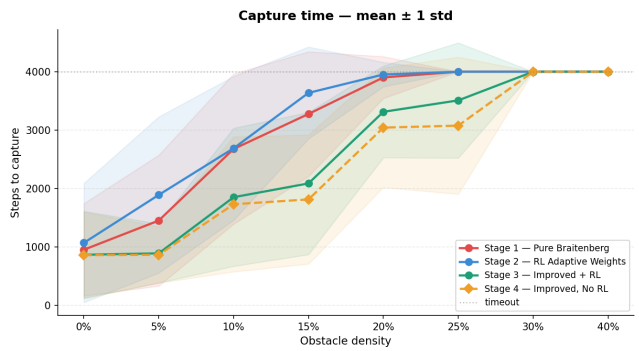

Figure 2: Mean capture time (steps) $\pm$ 1 std vs. obstacle density for all four pipeline stages. Dotted line: timeout ($T_{\max} = 4000$ steps). Stage 4 shown as a dashed line to distinguish it as the ablation condition.

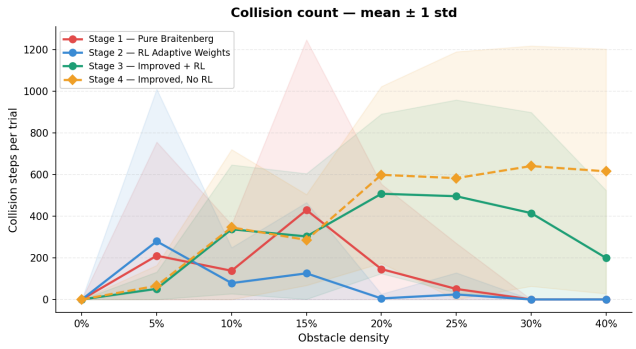

Figure 3: Mean collision count $\pm$ 1 std vs. obstacle density for all four pipeline stages. Stage 4 consistently exceeds Stage 3 at medium-to-high density, representing the sole measurable contribution of the RL collision penalty λ_c.

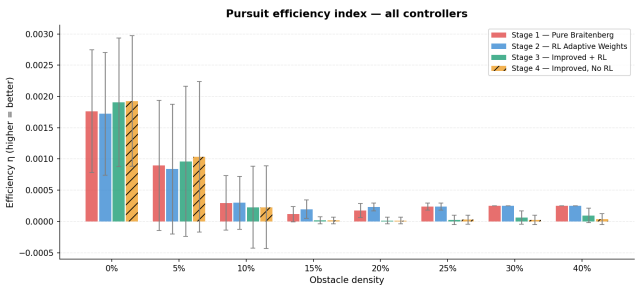

Figure 4: Pursuit efficiency index η for all four controllers. Stage 3 and Stage 4 lead at 0%–5% density but collapse to near zero from 10% onward as elevated collision counts dominate the denominator.

6. Discussion

6.1 Effect of Obstacle Density on Reactive Pursuit

The results demonstrate that obstacle density has a threshold effect on pursuit performance for the fixed-weight Braitenberg controller. Stage 1 maintains high capture rates (90–100%) at low density ($\rho \leq 5\%$) but undergoes rapid degradation between 10% and 20% — from 60% to 10% — with near-total failure by 25%. This sharp transition is consistent with prior work on reactive navigation in cluttered environments [8], where limit cycles and deadlock configurations emerge once obstacle spacing falls below the robot's effective turning radius and sensor lookahead distance. The 10%–20% range corresponds to real-world spaces transitioning from a lightly furnished office to a cluttered warehouse floor

— precisely the conditions under which a fixed reactive controller would be expected to fail.

6.2 Failure Mode Analysis

The collision count data reveals a non-monotonic pattern for Stage 1: counts rise from 210 steps at 5% to a peak of 429 steps at 15%, then fall sharply to 146 steps at 20% and near zero from 25% onward. This is the hallmark signature of a transition from *thrashing* to *stalling*. At 15% density, the robot is still mobile enough to attempt pursuit but encounters obstacles frequently; the avoidance channel fires repeatedly, producing wall contacts. Above 20%, the max-per-hemisphere aggregation clips both motor commands simultaneously to zero in dense pockets, pinning the robot in place. A stalled robot incurs no further collisions and generates no further captures, explaining both the collision drop and the success rate collapse at the same density threshold. This stalling failure mode is the primary motivation for the forward bias and motor floor introduced in Stage 3 and Stage 4.

6.3 Benefit of Adaptive Weight Tuning (Stage 2)

The Stage 2 results reveal that RL weight adaptation on the original Braitenberg base provides no meaningful improvement over Stage 1. At 5% density, Stage 2 achieved 80% capture vs. Stage 1's 90%, and consistently underperformed Stage 1 at 10% (65% vs. 60% — a marginal gain within noise), 15% (20% vs. 35%), and 20% (5% vs. 10%). The most likely explanation is that the REINFORCE-with-baseline algorithm, over 2,000 training episodes, converges to a near-zero output policy: with high-variance gradient estimates and a sparse capture reward, the safest strategy is to output $\Delta\alpha \approx \Delta\beta \approx 0$ and rely on the base controller. This is consistent with the near-identical pursuit efficiency curves for Stage 1 and Stage 2 (Figure 4). The finding highlights a fundamental limitation of meta-control: once the robot stalls for structural reasons, no weight adjustment can recover pursuit without changing the underlying kinematics.

6.4 Contribution of Improved Base Kinematics (Stage 3)

Stage 3 demonstrates that kinematic improvements combined with RL weight adaptation substantially extend the density range of successful pursuit. The 45-percentage-point advantage at 15% density (80% vs. 35% for Stage 1) and the non-zero success rates at 20% (55%) and 25% (20%) — where Stage 1 and Stage 2 have completely failed — confirm that the kinematic improvements are the primary driver. Mean capture times at 10%–15% (1,853 and 2,089 steps) are substantially faster than Stage 1 (2,679 and 3,277 steps), reflecting the robot's ability to continue navigating rather than stalling. The forward bias and centre-mean motor floor prevent stalling; cosine-weighted proximity aggregation produces smoother avoidance responses; and the stuck-recovery mechanism resolves deadlocks that the other mechanisms cannot.

The tradeoff is elevated collision counts: 337 steps at 10%, 303 at 15%, 508 at 20%, and 496 at 25% — all substantially higher than Stage 1 and Stage 2 at the same densities. The RL collision penalty ($\lambda_c = 0.15$) partially moderates this, as will be shown in the Stage 4 comparison.

6.5 Ablation: Does the RL Policy Add Value? (Stage 3 vs. Stage 4)

The Stage 3 vs. Stage 4 comparison directly answers whether RL weight adaptation contributes meaningfully when the kinematic improvements are already in place.

On capture rate, Stage 4 matches or exceeds Stage 3 at every density: both 100% at 0%–5%; Stage 4 higher at 10% (85% vs. 80%) and 15% (90% vs. 80%); equal at 20% (55%); Stage 4 higher at 25% (40% vs. 20%); both 0% at 30%–40%. Mean capture times for Stage 4 are also equal to or faster than Stage 3 at every density where captures occur (Figure 2). The kinematic improvements alone are therefore not only sufficient but actually yield better capture performance than the same improvements with RL weight adaptation.

The one metric where Stage 3 outperforms Stage 4 is collision count. Stage 3 incurs fewer collision steps than Stage 4 at every density

above 5%: 337 vs. 346 at 10%, 303 vs. 286 at 15% (Stage 3 marginally higher here), 508 vs. 599 at 20%, 496 vs. 583 at 25%, 415 vs. 641 at 30%, and 199 vs. 615 at 40%. The divergence is most pronounced at 30%–40%, where Stage 3's RL policy successfully suppresses obstacle contact at the cost of zero captures — the robot avoids collisions by barely moving. This confirms the RL collision penalty ($\lambda_c = 0.15$) is operating as designed, but the resulting behaviour is an overly cautious policy that sacrifices both capture rate and forward progress in exchange for reduced obstacle contact.

The practical implication is unambiguous: Stage 4 is the preferred controller for deployment where capture rate is the objective, achieving equal or better capture performance with no training overhead and fully deterministic, auditable behaviour. Stage 3 is preferable only when minimising physical obstacle contact is an independent constraint — for example, to limit wear on a real robot.

6.6 Interpretability and Practical Considerations

A key advantage of the Braitenberg framework is the interpretability of its behaviour: given the sensor readings and weights (α, β), the motor commands can be analytically derived with no hidden state. The hybrid extension partially retains this property — the Braitenberg topology remains fixed and the weight adjustments are bounded within $[-0.5, +0.5]$ of the nominal values by the tanh output scaling — but the learned policy π_θ introduces a degree of opacity that would require additional verification steps before deployment in safety-critical security contexts. The Stage 3 kinematic improvements (forward bias, motor floor, stuck recovery) are fully deterministic and rule-based, and therefore remain interpretable and auditable regardless of the learned weights.

7. Conclusion

This paper presented a quantitative evaluation of a four-stage Braitenberg-based pursuit controller pipeline in cluttered surveillance environments across obstacle densities ranging from 0% to 40%. The pure reactive controller (Stage 1) maintained 90–100% capture rate up to 5% obstacle density (mean capture times 955 and 1,454 steps), declining to 0% by 25%, with the primary failure mode identified as a transition from avoidance-channel thrashing at 15% (peak collision count 429 steps) to complete stalling above 20%. The Stage 2 hybrid learned-weight extension, applying REINFORCE-with-baseline weight adaptation to the original Braitenberg base, provided no improvement and in several cases degraded performance, confirming that parametric adaptation cannot overcome structural kinematic limitations. Stage 3, which combined cosine-weighted proximity blending, a forward bias, a mean-centred motor floor, and stuck recovery with RL weight adaptation, extended the operating range to 80% capture at 10%–15%, 55% at 20%, and 20% at 25%, at the cost of elevated collision counts (303–508 steps at 10%–25%). Critically, Stage 4 — the improved base with fixed weights and no RL — matched or exceeded Stage 3's capture rate at every density level, including 85% at 10% and 90% at 15% vs. Stage 3's 80%, while incurring moderately higher collisions at 30%–40% (615 vs. 199 steps). The RL policy's sole measurable contribution is collision reduction at high density, achieved by learning to be overly cautious rather than by improving pursuit. These results establish that the kinematic improvements are both necessary and sufficient for the performance gain, and that RL weight adaptation, as implemented here, provides no benefit to capture rate on the improved base.

The primary limitations of this work are the schematic 2D simulation (no sensor noise, actuator delay, or 3D terrain), the restriction to circular obstacles, and the use of a fixed bounce-mode intruder that does not model evasive behaviour. Future work should validate the architecture on a physical TurtleBot3 or equivalent platform, extend to non-circular obstacle geometries, and compare against a fully learned deep RL baseline with an evasive intruder.

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